JUNWEN MIAO

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EDUCATION

Carnegie Mellon University Aug 2026 - Dec 2027
Location: Pittsburgh, USA
M.Sc. in Machine Learning (Incoming)

Tongji University Sep 2022 - Jun 2026
Location: Shanghai, China
B.Eng. in Software Engineering; GPA: 4.75/5.0

University of California, Irvine Jun 2025 - Sep 2025
Location: Irvine, CA, USA
Exchange in Computer Science; GPA: 4.0/4.0

PUBLICATIONS

AgentIAD: Agentic Industrial Anomaly Detection via Adaptive Memory Augmentation
Junwen Miao*, Penghui Du*, Yingying Fan*, Yi Liu, Yu Wang, Runze He, Lida Huang, Yan Wang.
Under Review
Keywords: Visual Reasoning, Multimodal Agent, Industrial Anomaly Detection

ClawBench: Can AI Agents Complete Everyday Online Tasks?
Yuxuan Zhang, Yubo Wang, Yipeng Zhu, Penghui Du, Junwen Miao, Xuan Lu, Wendong XU, Songcheng Cai, Dongfu Jiang, Yi Lu, Yunzhuo Hao, Minyi Lei, Kai Zou, Huifeng Yin, Ping Nie, Liang Chen, Wenhu Chen, Kelsey R Allen.
Under Review
Keywords: AI Agents, Benchmark, Real-world Environment, LLM

RewardHarness: Self-Evolving Agentic Post-Training
Yuxuan Zhang*, Penghui Du*, Bo Li*, Cong Wei*, Junwen Miao, Huaisong Zhang, Songcheng Cai, Yubo Wang, Dongfu Jiang, Yuyu Zhang, Ping Nie, Wenhu Chen, Changqian Yu, Kelsey R Allen.
Under Review
Keywords: Reward Model, Harness Engineering, Image Editing

SportR: A Benchmark for Multimodal Large Language Model Reasoning in Sports
Haotian Xia*, Haonan Ge*, Junbo Zou*, Hyun Woo Choi, Xuebin Zhang, Danny Suradja, Botao Rui, Ethan Tran, Wendy Jin, Zhen Ye, Xiyang Lin, Christopher Lai, Shengjie Zhang, Junwen Miao, Shichao Chen, Rhys Tracy, Vicente Ordonez, Weining Shen, Hanjie Chen.
ICLR, 2026
Keywords: Multimodal Large Language Models, Sports Understanding, Benchmark

RESEARCH EXPERIENCE

AIR, Tsinghua University Apr 2025 - Present
Research Intern (Remote); Advisor: Yan Wang *Beijing, China*

- Led AgentIAD, formulating industrial anomaly detection as a sequential inspection problem where the model actively acquires local, comparative, and external evidence.
- Designed tool-aware SFT trajectories and GRPO rewards for adaptive tool use, visual memory, and retrieval memory.
- Worked on embodied open-vocabulary detection, including affordance-aware localization, language-conditioned grounding, and task-relevant object perception.

University of Waterloo Mar 2026 - May 2026
Research Intern (Remote); Advisor: Wenhu Chen *Waterloo, Canada*

- Built execution-based evaluation tasks for web agents on real websites, with success measured by final states and HTTP-level outcomes.
- Recorded interaction traces including screenshots, browser actions, agent messages, and network requests for reproducible failure analysis.
- Built harness-based post-training settings where agents improve from execution feedback, reward checks, and generated trajectories.

University of California, Irvine Jun 2025 – Sep 2025
Research Intern (On-Site); Advisor: Weining Shen *Irvine, CA*

- Worked on table-tennis task annotation for SportR, covering rule understanding, event judgment, scoring logic, and fine-grained visual evidence.
- Analyzed model performance on table-tennis reasoning tasks, with attention to grounding errors, ambiguous ball trajectories, and rule-specific failure cases.

INTERNSHIP EXPERIENCE

Embodied Research Centre, Agibot Mar 2026 - Present
Algorithm Engineering Intern; Mentors: Yi Liu, Jianlan Luo *Shanghai, China*

- Worked on VLA pretraining with large-scale embodied and vqa data, covering data processing, model training, and high-level evaluation.
- Implemented inference pipelines for Qwen3.5-based models, VLA models, and FAST-augmented VLA models to support downstream testing and open-loop evaluation.

HONORS AND AWARDS

Outstanding Graduate
Award for top undergraduate graduates at Tongji University 2026
Outstanding Student Scholarship
Award for top students at Tongji University 2023, 2024, 2025
Outstanding Undergraduate Student
Award for top 1% of undergraduate students 2024
National Undergraduate Mathematical Contest in Modeling
Third Prize in Shanghai, China 2023